Machine Learning

1. Define Machine Learning in your own words.
Explain how it is different from traditional rule-based programming with a real-world example.

2. What are **Independent Variables** and **Dependent Variables** in Machine Learning?
Using the Ball classification case study, clearly identify:

- Independent variables
- Dependent variable

3. Explain the difference between Features and Labels.
Are features always numeric? Justify your answer.

4. What is Supervised Machine Learning?
Why is labeled data mandatory in supervised learning?

5. Explain why we cannot train and test a model using the same dataset.
What problem may occur if we do so?

6. What is the purpose of splitting a dataset into training and testing sets?
What is the typical ratio used in industry?

7. Explain the meaning of the following variables:

- X_train
- X_test
- y_train
- y_test

8. What is the role of the parameter test_size=0.2 in train_test_split()?
What happens if we change it to 0.5?

9. What is the purpose of random_state=42 in train_test_split()?
What will happen if we do not provide it?

10. Explain step-by-step what happens internally when we execute:

model.fit(X_train, y_train)

Machine Learning

1. Differentiate between:

- model.fit()
- model.predict()

Explain which one is used during training and which during testing.

2. Define Accuracy.
Write the mathematical formula for accuracy and explain it in simple words.

3. If a model predicts 80 correct results out of 100 samples:

- What is the accuracy?
- Is high accuracy always good? Explain with reasoning.

4. Explain the concept of a Confusion Matrix.
Why is it more informative than accuracy?

5. In the Ball classification case study (Cricket vs Tennis):

Explain clearly:

- True Positive
- True Negative
- False Positive
- False Negative

Give real meaning using the ball example.

6. If a tennis ball is predicted as cricket ball:

- What type of error is this?
- Why is it called that?

7. In a confusion matrix:

	Predicted Cricket	Predicted Tennis
Actual Cricket	40	5
Actual Tennis	3	52

Calculate:

- Accuracy
- Total Errors
- Total Correct Predictions

8. What is the difference between:

- Binary Classification
- Multiclass Classification

Give one real-world example for each.

9. Explain how confusion matrix changes in multiclass classification compared to binary classification.

10. In the Iris dataset:

- What are the features?
- What is the label?
- Why is it a multiclass classification problem?
- How many possible output classes

Machine Learning

1. Define Artificial Intelligence (AI).
How is AI different from traditional software systems?

2. Explain the three types of AI based on capability:

- Narrow AI
- General AI
- Super AI

3. Is Machine Learning a part of AI or is AI a part of Machine Learning?
Explain their relationship using a hierarchy diagram in your notebook.

4. Define Machine Learning.
Explain the three main types of Machine Learning with one example each:

- Supervised Learning
- Unsupervised Learning
- Reinforcement Learning

5. Differentiate clearly between:

- Supervised Learning
- Unsupervised Learning

Based on:

- Data requirement
- Output
- Real-world applications

6. What is Reinforcement Learning?

7. Define Deep Learning.
How is Deep Learning different from traditional Machine Learning?

8. What is Data Science?
Explain how Data Science is different from Machine Learning.

9. Explain the complete Data Science lifecycle:

- Data Collection
- Data Cleaning
- Data Analysis
- Model Building
- Deployment

10 What is Generative AI (GenAI)?
How is it different from traditional predictive Machine Learning models?